

Air transport of brachycephalic dogs: physiological risks, risk factors and regulatory framework

Jessica Ysabel Camacho Garcia, M.V.Z., CMVP 12434 — ORCID 0009-0002-6837-5311

Zoovet Travel, Trujillo, Peru

Victor Jesus Camacho Paz, M.V., CMVP 3103

GERESA La Libertad / MINSA, Trujillo, Peru

Technical Series Vol. III · Publisher: Zoovet Travel - zoovettravel.com

Trujillo, La Libertad, Peru · May 2026 · DOI: 10.5281/zenodo.20370359

Abstract

Objective. To analyze the specific physiological risks of air transport in brachycephalic dogs, the individual factors that modulate risk, and the regulatory framework from airlines and international institutions. **Method.** Technical review of brachycephalic obstructive airway syndrome (BOAS) anatomy, respiratory physiology in the aeronautical environment (cabin pressure, PO₂, thermoregulation and panting), comparative cabin vs hold analysis, major airline operational policies (2024-2025), IATA LAR regulations, DOT animal incident reports and FAA provisions. **Results.** BOAS generates a severity continuum that interacts with cabin hypobaric hypoxia (equivalent to 6,000-8,000 ft), heat and confinement stress; panting, the primary thermoregulatory mechanism, is itself dependent on the compromised airway. Individual risk factors include BOAS grade, body condition, age and ambient temperature. Airline policies vary substantially, with most prohibitions applying to the hold, not the cabin. **Conclusion.** Risk in brachycephalic dogs is not binary but graded; pre-flight veterinary assessment with BOAS grade certification is the primary clinical determinant of the travel decision.

Keywords: brachycephalic, BOAS, air transport, hypobaric hypoxia, thermoregulation, panting, IATA, DOT, FAA, pre-flight assessment

Abstract

1. Introduction: beyond the prohibition

Over the last fifteen years, veterinary consultations related to the air transport of brachycephalic breeds have increased in parallel with the popularity of these breeds. According to O'Neill et al. (2018), the French Bulldog moved from a marginal breed in the United Kingdom to one of the most registered, with all that this implies for the international mobility demands of their owners.

The airline industry response has been gradual but sustained: progressive hold restrictions first; in some cases extension of limitations to the cabin thereafter. The declared rationale in every case is the same: the greater respiratory vulnerability of these breeds to the conditions of the aeronautical environment. That vulnerability has a genuine physiological basis, which is the central subject of this review.

The correct question is not "can brachycephalic dogs fly?" but "under what conditions, with what level of risk, and in which specific dog?" Answering that question with rigour requires first understanding why these animals represent a group with differential physiology in the context of air transport.

2. Brachycephalic anatomy and obstructive syndrome (BOAS)

Pathophysiological basis — peer-reviewed published science

2.1. The structural problem: shortened bone, unreduced soft tissue

In clinical practice, brachycephalic patients with signs consistent with BOAS present a variable combination of upper airway resistance and dynamic airway compromise that cannot be understood without

grasping its anatomical origin. The shortening of the facial skeleton, artificially selected over decades of breeding, has reduced the bony scaffold but not the soft tissues that occupy it: the nasal mucosa, soft palate, tongue and tonsils remain in a space that can no longer accommodate them (Liu et al., 2017).

The result is a multilevel obstruction documented with precision in the literature: stenotic nares, aberrant intranasal turbinates, elongated and hypertrophic soft palate, relative macroglossia and, in the most advanced cases, secondary laryngeal collapse (Krainer & Dupré, 2022). Liu et al. (2016) quantified using whole-body barometric plethysmography that nasal and oropharyngeal respiratory resistance in brachycephalic dogs is significantly greater than in mesocephalic breeds, even at rest and without physical exertion.

Figure 1. Anatomical components of BOAS — from conformation to functional obstruction (synthesis from published literature)

Reduction of the nasal aperture. Increases resistance to airflow at the first point of entry. Correlates with functional BOAS score in field studies (Lilja-Maula et al., 2024).

Intranasal bony growths that partially occlude the nasal passage. Frequent in Pugs and Bulldogs. Documented by CT in multiple case series (Krainer & Dupré, 2022).

The most studied component. Eivers et al. (2023) described histological changes in the soft palate tissue of brachycephalic dogs including muscle hypertrophy and oedema, contributing to dynamic obstruction during inspiration.

Chronic consequence of sustained intra-laryngeal negative pressure. Represents the final progression stage of BOAS. In these cases the surgical prognosis is more guarded (Krainer & Dupré, 2022).

2.2. BOAS as a severity continuum, not a binary diagnosis

Packer et al. (2015) established that the risk of BOAS increases continuously with the degree of brachycephaly, measured through the craniofacial ratio (CFR). In their study — considered a methodological reference in the field — the probability of functional BOAS exceeded 50% in dogs with a CFR below 0.20, a value characteristic of extreme Pugs and Bulldogs.

This progressiveness has direct implications for assessing flight risk: not all brachycephalic dogs have the same level of respiratory compromise. A Boston Terrier with wide nares and a non-elongated palate has a very different risk profile from a Pug with grade III BOAS. Airline policies, however, do not discriminate by individual severity: they classify by breed, introducing clinically significant variability that can only be resolved through individual veterinary assessment.

Rigas et al. (2024), in a cohort of more than 14,000 French Bulldogs, Pugs and English Bulldogs under primary veterinary care in the United Kingdom, documented that the prevalence of signs compatible with BOAS varies considerably by breed and diagnostic methodology. This variability reaffirms the impossibility of assigning a uniform risk level to all dogs of a given breed.

3. Respiratory physiology in the aeronautical environment

3.1. Cabin pressurisation and reduced PO₂

Commercial aircraft maintain a cabin pressure equivalent to an altitude of approximately 1,800 to 2,400 metres above sea level (approximately 565–750 mmHg), representing a reduction in the partial pressure of oxygen of the order of 15–20% relative to sea level. In a healthy normocephalic dog, this level of relative hypoxia is well tolerated through minor ventilatory adjustments. In a brachycephalic dog with pre-existing multilevel obstruction, the same hypoxia adds to an already compromised system.

Published evidence in human medicine has documented the effects of moderate altitude hypoxia in patients with upper airway obstruction; however, direct extrapolation of these findings to the canine species is not validated in the available veterinary literature. Specific data on the physiological response of brachycephalic dogs to cabin hypoxia under real aeronautical conditions are scarce. What is documented in

veterinary studies is that animals with moderate to severe BOAS present increased upper airway resistance and ventilatory compromise even at rest and at sea level (Liu et al., 2016; Mitze et al., 2022). If a reduction in ambient PO₂ is added to that compromised baseline state, greater ventilatory demand would be expected on general physiological grounds; however, the clinical magnitude of that effect in flight has not been quantified in controlled studies with the species.

3.2. Thermoregulation and panting as a dependent mechanism

Dogs lack efficient sweat glands for dissipating body heat. Their primary thermoregulatory mechanism is panting — a process that depends on ventilation of the upper airway mucosa to facilitate evaporation and heat exchange. Davis, Cummings & Payton (2017) demonstrated that brachycephalic dogs are significantly less efficient at thermoregulation through this mechanism, and that the combination of brachycephaly and overweight further increases the difficulty.

In the context of air transport, this is relevant for two reasons. First, hold temperatures are not uniformly controlled during all phases of the flight, including loading and runway waiting. Second, the stress of transport per se increases metabolic heat production, demanding greater thermoregulatory efficiency precisely when the system is least able to provide it.

O'Neill, James et al. (2020), in an epidemiological study on heat-related illness episodes in dogs under primary veterinary care in the United Kingdom, identified over-representation of brachycephalic breeds — particularly English Bulldogs, French Bulldogs and Pugs — among documented cases. This study was conducted under terrestrial ambient conditions and does not address the aeronautical environment. Its relevance to the present analysis is indirect: the data suggest that these animals present reduced tolerance to thermoregulatory demands under heat conditions, which constitutes a plausible antecedent but not direct evidence of greater risk in the specific context of air transport.

3.3. Stress and oxygen consumption

Transport-associated stress — separation from the owner, unfamiliar environment, noise, vibration, confinement — activates the hypothalamic-pituitary-adrenal axis and the sympathetic nervous system, with documented effects on heart rate, respiratory rate and tissue oxygen consumption. In a brachycephalic dog with diminished functional ventilatory reserve, the increase in metabolic demand induced by stress sharpens the discrepancy between oxygen supply capacity and tissue demand.

In a brachycephalic dog with pre-existing multilevel obstruction, the increase in metabolic demand induced by stress sharpens the discrepancy between oxygen supply capacity and tissue demand. Mitze et al. (2022), in their review of published clinical evidence on BOAS as a systemic problem, note that animals with moderate to severe forms may present signs consistent with chronic hypoxia; the authors reference prior primary evidence on haematological alterations in brachycephalic dogs, though the original studies quantifying these alterations are not within the scope of that review and exceed the bibliographic scope of this document. The assertion of chronic baseline hypoxia and elevated haematocrit as a consistent finding in the brachycephalic population should be understood, in the context of this analysis, as a description of what has been reported in recent clinical reviews, not as a primary datum verified in this document.

4. Cabin vs hold: comparative technical analysis

The distinction between transport in the passenger cabin and transport in the cargo hold is the central axis of airline policies regarding brachycephalic dogs, and has technical justification. It is not purely an administrative distinction.

In the cabin, the animal travels in a carrier placed under the owner's seat, in a temperature-regulated environment and in direct presence of the responsible party, who can detect signs of respiratory distress in real time and intervene (alert crew, reposition the animal, open the carrier if the situation requires it). This capacity for continuous supervision is qualitatively different from the hold situation.

The United States Department of Transportation (DOT, 2024) animal incident reports, publicly accessible since 2005, document losses, injuries and deaths of animals during air transport on US carriers. Their review has been noted in the specialist literature as the instrument that motivated policy changes at North American airlines during the 2010–2020 decade regarding brachycephalic breeds. The raw data from these reports are available at [transportation.gov/airconsumer](https://www.transportation.gov/airconsumer); their detailed quantitative analysis exceeds the scope of this document and requires direct access to primary records for any claim of prevalence or proportion.

5. Regulatory framework and airline operational policies

5.1. IATA LAR Regulations

The IATA Live Animals Regulations (LAR, 2024) establish the international technical framework for the transport of animals on commercial aircraft, including specifications for containers, ventilation and minimum dimensions. The LAR constitute the technical basis on which each airline builds its specific operational policy. They do not themselves prohibit the transport of brachycephalic breeds, but do establish container requirements — including the requirement for additional space for these breeds in airlines that still accept them in the hold — that reflect the need to maximise ventilation (IATA, 2024).

5.2. US Department of Transportation Animal Incident Reports

Since 2005, US regulation (49 U.S.C. §41721; 14 C.F.R. Part 235) has required US commercial airlines operating aircraft with more than 60 seats to file monthly reports on incidents involving the loss, injury or death of animals during transport. These data are public and accessible on the DOT portal ([transportation.gov](https://www.transportation.gov)). Their systematic review was the instrument that allowed the over-representation of brachycephalic breeds in adverse events to be identified, motivating policy changes at North American airlines during the 2010–2020 decade (DOT, 2024).

5.3. FAA and cabin conditions

The FAA does not establish breed-specific restrictions for in-cabin transport on aircraft, but does regulate the general safety requirements applicable to pet containers and their stowage under the seat (FAA, 2023). Cabin temperature and pressurisation conditions are regulated by the FAA as part of the airworthiness standards of commercial aircraft.

5.4. Comparative table of major airline policies (2024–2025)

The global pattern is consistent: the aviation industry trend is to prohibit hold transport of brachycephalic breeds and permit cabin transport under size and weight conditions, with notable exceptions (Emirates) that prohibit it in both compartments. This trend responds to accumulated incident data, not to arbitrary preference.

6. Individual risk-modifying factors

Given that BOAS is a continuum and not all brachycephalic breeds carry the same risk profile, evaluation of individual factors is essential to contextualise the risk of a specific flight for a specific animal.

The clinically weightiest factor. An animal with functional BOAS grade I has a very different profile from one with grade III with laryngeal collapse. Assessment using functional grading systems — such as that described by Liu et al. (2016) using plethysmography — allows severity to be objectified, though it is not always available in primary care settings.

Davis et al. (2017) documented that excess weight significantly increases thermoregulatory difficulty and respiratory burden in brachycephalic dogs. Liu et al. (2017) confirmed the association between higher body condition score and greater functional BOAS severity. An obese brachycephalic dog has a consistently higher flight risk than one at ideal weight.

Young animals — under 1 year — have airways still developing; geriatric animals may have accumulated secondary BOAS lesions (laryngeal collapse, systemic hypertension) that worsen the risk profile. Airlines establish minimum transport ages (generally 8–16 weeks) for reasons of physiological maturity.

Cumulative exposure to stress factors and reduced PO₂ increases with flight duration. Flights with stopovers present additional periods of loading, runway waiting and changing environmental conditions, especially during hold handling.

Hall, Carter & O'Neill (2022) documented that heat-related illness episodes in high-risk breeds show marked seasonality, with peaks during months of higher temperature. Tarmac heat during loading and pre-departure waiting represents an additional risk factor in summer or on tropical routes.

Animals that have undergone corrective BOAS surgery (rhinoplasty, palatoplasty) may present an improved risk profile relative to their pre-operative state, although the literature does not allow precise quantification of the magnitude of this improvement in the specific context of air transport.

7. Pre-flight veterinary assessment

Description of practices — not prescriptive or individualised

Pre-flight veterinary assessment of a brachycephalic dog aims to document the animal's baseline respiratory status, identify modifiable risk factors and provide the owner with the information needed for an informed decision. It is not an administrative formality.

In published clinical practice, pre-flight assessment of brachycephalic animals has included evaluation of resting respiratory rate and effort, auscultation of the upper airway and thorax, evaluation of body condition and weight, and review of the clinical history for previous episodes of respiratory distress, syncope or BOAS signs. Lilja-Maula et al. (2017) described a clinical grading system that in their study context was supplemented with exercise tests to objectify the functional severity of BOAS. The description of these components reflects what is documented in the literature; their application in each specific case is the exclusive decision of the responsible veterinarian and cannot be derived from this document.

Some airlines and organisations such as IPATA have developed Brachycephalic fit-to-fly assessments, the objective of which is to discriminate, within the "brachycephalic breed" category, individuals with severe BOAS that contraindicate flight from those with mild-to-moderate compromise in which risk can be managed. These assessments have not been universally adopted by airlines.

8. Limitations of this analysis

- Data on physiological outcomes of brachycephalic dogs under real aeronautical conditions are scarce in the published veterinary literature. Most available evidence comes from clinical studies on the ground and from pathophysiological extrapolation, not from controlled studies aboard aircraft.
- DOT incident data have capture limitations (US airlines with more than 60 seats only; not all situations are reportable; assignment of cause of death may be imprecise) and should not be extrapolated as universal mortality rates.
- The airline policies described in Section 5 reflect the documented state as of 2024–2025 and may have changed. All information must be verified directly with the airline before booking.
- Individual variability within each brachycephalic breed is substantial. This document cannot and does not intend to replace individualised veterinary assessment of each specific animal.
- Technical hold conditions vary between aircraft types, airlines and routes. There is no single verifiable standard of hold temperature or ventilation for all commercial aircraft in operation.

9. Technical conclusion

Brachycephalic dogs present a physiology that can be compromised by the conditions of the aeronautical environment — reduced PO₂, variable temperature, stress and confinement — proportionally more so than normocephalic breeds. This differential risk is not uniform within the group: it depends on individual BOAS severity, body condition, age, flight duration and the capacity for supervision during transport.

The global airline trend — prohibiting hold transport and permitting cabin transport under controlled conditions — reflects precisely this risk asymmetry between the two compartments: the cabin offers direct supervision and regulated temperature; the hold does not. This is not a perfect response to the problem, but it is one with coherence with the available evidence.

The scientific literature does not contain a well-founded prohibition on air transport of all brachycephalic breeds under all conditions. What does exist is an evidence base that justifies differentiated caution, individual assessment and a systematic approach that considers the specific animal, airline, route and travel conditions.

References

- Davis, M. S., Cummings, S. L., & Payton, M. E. (2017). Effect of brachycephaly and body condition score on respiratory thermoregulation of healthy dogs. *Journal of the American Veterinary Medical Association*, 251 (10), 1160–1165. <https://doi.org/10.2460/javma.251.10.1160>
- Eivers, N., Liuti, T., Ladlow, J., & Engeham, J. (2023). Characterization of the soft palate in brachycephalic dogs with brachycephalic obstructive airway syndrome. *Veterinary Surgery*, 52 (4). <https://doi.org/10.1111/vsu.13926>
- Federal Aviation Administration (FAA). (2023). Air travel for pets . Federal Aviation Administration. https://www.faa.gov/travelers/fly_pets
- Hall, E. J., Carter, A. J., & O'Neill, D. G. (2022). Describing the evolution of heat-related illness in dogs present to UK primary care veterinary practices: 10-year study. *Scientific Reports*, 12 , 18025. <https://doi.org/10.1038/s41598-022-22849-0>
- International Air Transport Association (IATA). (2024). Live Animals Regulations (51st ed.). IATA. <https://www.iata.org/en/publications/store/live-animals-regulations/>
- Krainer, D., & Dupré, G. (2022). Brachycephalic obstructive airway syndrome. *Veterinary Clinics of North America: Small Animal Practice*, 52 (3), 749–780. <https://doi.org/10.1016/j.cvsm.2022.01.013>
- Lilja-Maula, L. I. O., Mäki, K. H., Aromaa, M. K., & Rajamäki, M. M. (2024). Evaluation of brachycephalic obstructive airway syndrome breeding test results in Finland from 2017 to 2022. *Acta Veterinaria Scandinavica*, 66 (1), 35. <https://doi.org/10.1186/s13028-024-00755-9>
- Lilja-Maula, L., Lappalainen, A. K., Hyytiäinen, H. K., Kuusela, E., Kere, J., & Rajamäki, M. M. (2017). Comparison of barometric whole-body plethysmography and a clinical respiratory grading system for diagnosis of brachycephalic obstructive airway syndrome in French bulldogs. *BMC Veterinary Research*, 13 , 181. <https://doi.org/10.1186/s12917-017-1275-z>
- Liu, N-C., Adams, V. J., Kalmar, L., Ladlow, J. F., & Sargan, D. R. (2016). Whole-body barometric plethysmography characterizes upper airway obstruction in 3 brachycephalic breeds of dogs. *Journal of Veterinary Internal Medicine*, 30 (3), 853–865. <https://doi.org/10.1111/jvim.13933>
- Liu, N-C., Troconis, E. L., Kalmar, L., Price, D. J., Wright, H. E., Adams, V. J., Sargan, D. R., & Ladlow, J. F. (2017). Conformational risk factors of brachycephalic obstructive airway syndrome (BOAS) in pugs, French bulldogs, and bulldogs. *PLOS ONE*, 12 (8), e0181928. <https://doi.org/10.1371/journal.pone.0181928>
- Mitze, S., Barrs, V. R., Beatty, J. A., Hobi, S., & Błczkowski, P. M. (2022). Brachycephalic obstructive airway syndrome: much more than a surgical problem. *Veterinary Quarterly*, 42 (1), 213–223. <https://doi.org/10.1080/01652176.2022.2145621>
- O'Neill, D. G., James, H., Brodbelt, D. C., Church, D. B., & Pegram, C. (2020). Epidemiology of brachycephalic obstructive airway syndrome in dogs under primary veterinary care in the UK. *Scientific Reports*, 10 , 9785. <https://doi.org/10.1038/s41598-020-66015-8>
- O'Neill, D. G., James, H., Brodbelt, D. C., Church, D. B., & Pegram, C. (2018). Demography and health of French bulldogs under primary veterinary care in the UK in 2013. *PLOS ONE*, 13 (4), e0196284. <https://doi.org/10.1371/journal.pone.0196284>
- Packer, R. M. A., Hendricks, A., Tivers, M. S., & Burn, C. C. (2015). Impact of facial conformation on canine health: Brachycephalic obstructive airway syndrome. *PLOS ONE*, 10 (10), e0137496. <https://doi.org/10.1371/journal.pone.0137496>
- Rigas, I., O'Neill, D. G., Church, D. B., Sargan, D. R., Ladlow, J. F., & Liu, N-C. (2024). Brachycephalic obstructive airway syndrome (BOAS) in French Bulldogs, Pugs and Bulldogs in the UK: Prevalence and risk factors. *Veterinary Record*, 194 (6). doi:10.1002/vetr.4144
- U.S. Department of Transportation (DOT). (2024). Animal incident reports . Office of Aviation Consumer Protection. <https://www.transportation.gov/airconsumer/animal-incident-reports>